

Estimating the distribution of a random variable



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Abstract

Methods for reporting scientific results so that other researchers can reproduce them are discussed. To illustrate how these methods work, we generate means and histograms by sampling Bernoulli random variables. We first show how we can use resampling to calculate error bars on the estimates of the expectations we get using this procedure. We then show how the central limit theorem can be used to reduce the cost of this procedure.

1 Introduction

Reporting the results from experiments and simulations that generate random variables is difficult. If we are doing our simulations and experiments properly, it should be possible for other people to reproduce our results. The particular set of values that we obtain from an experiment or simulation will often be subject to factors beyond our control, so there will be a degree of randomness in our results. We will thus often report information on the distribution of results that we obtain rather than the values. Other researchers can then test whether they have performed the same experiment as us by testing whether or not the distribution of results they obtain in their investigation is the same as the distribution of results we got.

In this report, I will discuss the tools that we use to obtain and report information on the distribution of results. I will begin in section 2 by discussing how we can include error bars on an estimate of a sample mean by calculating the sample mean multiple times. In section 3 I will then discuss how we can extract an estimate of the probability mass function and how suitable error bars on this estimate can be obtained by resampling. The report will then finish in section 4 by discussing how we can use the insight that is provided by the central limit theorem to report these errors more cheaply. Throughout the report, I will demonstrate the methods by generating multiple Bernoulli random variables. The code I have used for generating these Bernoulli random variables is shown below:

```
1 import numpy as np
2
3 def bernoulli(p) :
4     if np.random.uniform(0,1)<p : return 1
5     return 0
```

Listing 1: Generating Bernoulli random variables

2 Reporting the mean

The mean for a sample of n identical random variables, X_i can be calculated as follows:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

We can write a Python function to generate sample means by adding together n random variables as follows:

```
1 # Function takes parameter of random variable (p) and number of random
   variables to add together (n) as input
2 def sample_mean( p, n ) :
3     mean = 0
4     for i in range(n) : mean = mean + bernoulli(p)
5     return mean / n
```

Listing 2: Generating sample means

The mean that is calculated using the formula and function above should converge on the expectation of the random variable $E(X)$ as n is increased. Importantly, however, the mean is not equal to the expectation of the random variable. The mean is calculated by adding together random variables. **The mean is thus a random variable.** When we report a mean, we, therefore, need to provide information on the distribution that was sampled and not simply the value of the mean.

To give a sense of how this is done, I generated 100 sample means using the formula above. Each of these sample means was calculated by adding 200 Bernoulli random variables with a p parameter of 0.5. We would thus expect the expectation to be 0.5. The median of my 100 estimates of the sample mean was 0.505. 5% of the sampled values were less than or equal to 0.445. 95% of the sampled values were less than or equal to 0.565. Another person repeating this experiment would have a 90% chance of obtaining a mean between 0.445 and 0.565. Furthermore, the range of values identified brackets the true value for the expectation of 0.5.

3 Estimating the histogram

Figure 1 shows an estimate of the probability mass function for a Bernoulli random variable with $p = 0.3$. 200 Bernoulli random variables were generated and the fraction of times each of the two possible outcomes appeared was reported using the Python function shown below:

```
1 def genhisto( nsamples, nbins, p ) :
2     counts = np.zeros(nbins)
3     for i in range(nsamples) :
4         xbin = int( bernoulli(p) )
5         counts[xbin] = counts[xbin] + 1
```

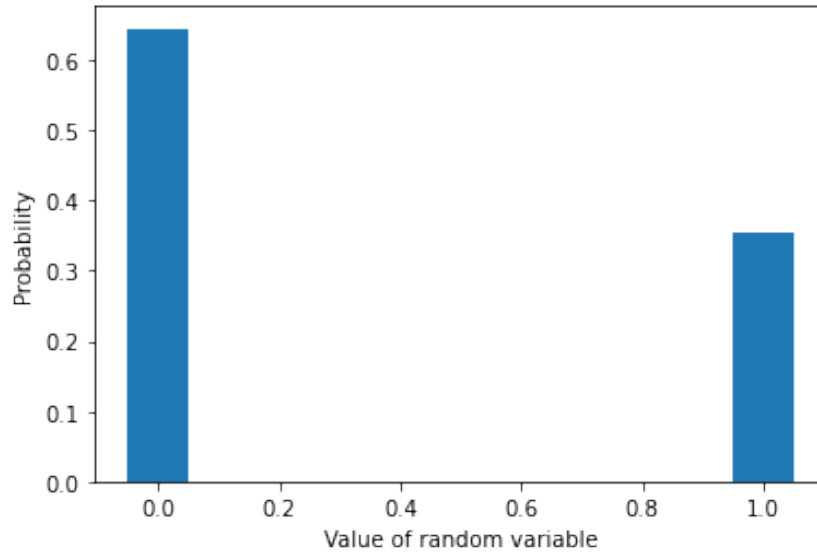


Fig. 1. An estimate of the probability mass function for the Bernoulli random variable that was calculated by sampling.

```
6 return counts / nsamples
```

Listing 3: Generating histograms

The heights of the bars in figure 1 are random variables. We must, therefore, provide suitable error bars on these estimates of the probabilities if we wish to make our result reproducible.

We can obtain error bars on the heights of the bars in a histogram by using the resampling technique that was introduced in section 2. This procedure was used to generate the error bars that are illustrated in figure 2. The error bars in this figure are centred on the median that was obtained by calculating 200 resamples of the histogram. Each of these estimates of the histogram was generated by sampling 200 Bernoulli random variables with $p = 0.3$. The error bar then shows the 90% confidence limit around this estimate of the probability. In figure 2 the probability mass function that was sampled to generate the estimate of the distribution is shown using the solid blue bars. You can see that the actual value of the probability is within the 90% confidence limit of the estimate generated by sampling. The median is not equal to the actual value of the probability, however.

4 The central limit theorem

The methods that we have introduced in previous sections for calculating errors are expensive. The expense of these methods arises because we have to generate multiple instances of the sample mean. We can avoid this expense by recalling the following results for the expectation and variance of the sample mean:

$$\mathbb{E}(\bar{X}) = \mathbb{E}(X) \quad \text{and} \quad \text{var}(\bar{X}) = \frac{\text{var}(X)}{n}$$

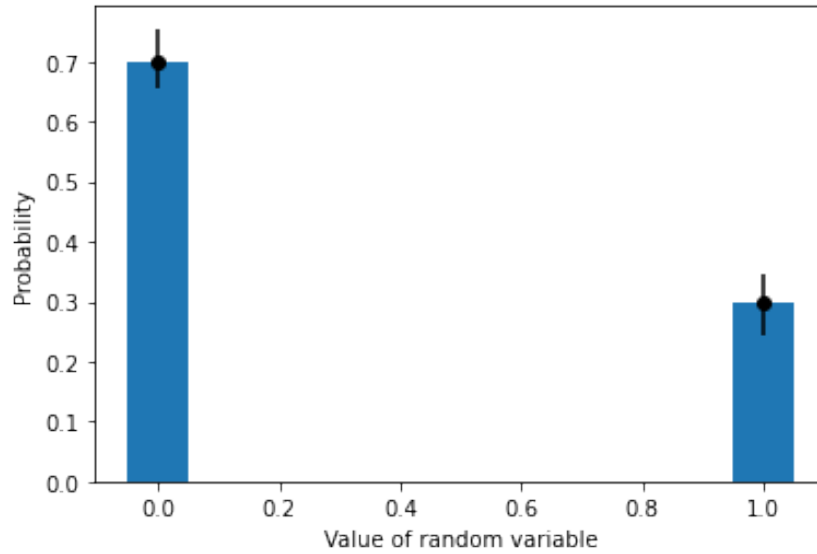


Fig. 2. The black error bars show an estimate of the probability mass function for the Bernoulli random variable that was calculated by sampling. The error bars illustrate the 90% confidence limit on this estimate of the distribution. The blue bars show the probability mass function for the distribution that was sampled.

In these expressions the sample mean $\bar{X}$ is calculated by adding together n identical and independent random variables X with expectation $\mathbb{E}(X)$ and variance $\text{var}(X)$.

We can proceed further by remembering that the central limit theorem tells us that sample means are approximately normally distributed. We thus do not need to sample the mean multiple times as we know that the means is a sample from a normal distribution with expectation and variance given by the expressions above.

If we have computed a sample mean from n random variables, X_i , we can thus estimate the sample variance from these variables by using:

$$S^2 = \frac{n}{n-1} \left[\frac{1}{n} \sum_{i=1}^n X_i^2 - \left(\frac{1}{n} \sum_{i=1}^n X_i \right)^2 \right]$$

We can then use the inverse of the cumulative probability distribution for the standard normal distribution, $\Phi^{-1}(p)$, as follows to calculate the value, x , that a sample mean has a probability of p of being less than or equal to:

$$x = \sqrt{\frac{S^2}{n}} \Phi^{-1}(p) + \mu$$

We can thus generate 200 Bernoulli random variables with $p = 0.5$ as opposed to the 20,000 random variables we generated in section 2. We can estimate the sample mean and sample variance from these random variables as 0.470 and 0.250, respectively. By inserting

these values into the expressions above, we can then say that a researcher who repeats our experiment has a 90% chance of obtaining a value between 0.412 and 0.528.

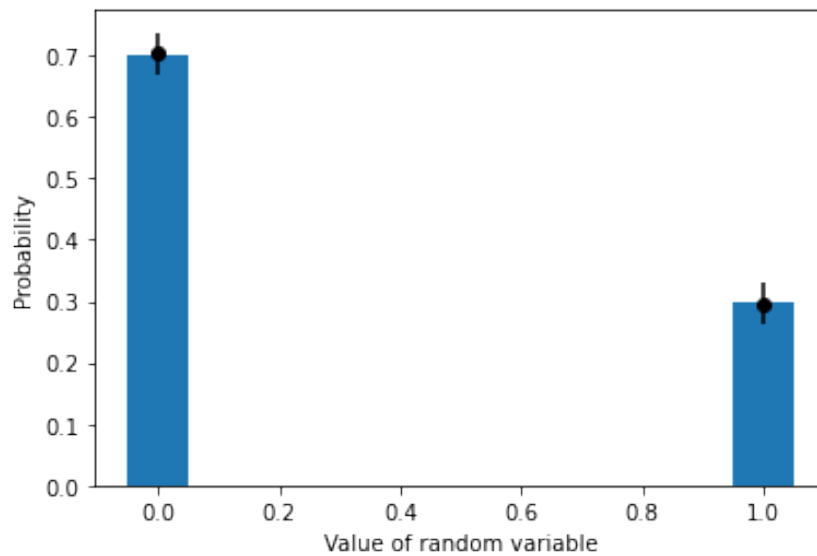


Fig. 3. The black error bars show an estimate of the probability mass function for the Bernoulli random variable that was calculated by sampling. The error bars illustrate the 90% confidence limit on this estimate of the distribution. These errors were calculated by assuming the mean is normally distributed. The blue bars show the probability mass function for the distribution that was sampled.

Notice that we can also use these ideas based on the central limit theorem when we estimate the histogram for the distribution. Figure 3 shows an estimate of the histogram for a Bernoulli random variable with $p = 0.3$ that was obtained in this way. We generated 500 Bernoulli random variables to generate the estimates of the probabilities that are shown by the black error bars in figure 3. The error bars in the figure show the 90% confidence limit on these estimates of the probabilities. The blue bars then show the probability mass function that was sampled to generate these estimates. You can see that the actual probabilities are within the error bars on the estimates. Once again, though, the estimate of the distribution is not equal to the actual probability mass function.